

IE340 Series

Industrial Ethernet Layer 3 Switches

Overview

IE340 Series access switches provide network infrastructure to digitally transform industry.

Allied Telesis IE340 Series switches are the perfect solution for access connectivity in industrial automation and control networks.

Their low latency, high availability and high capacity enables seamless integration with industrial applications, supporting multiple transmission streams. They are hardened to withstand difficult environmental conditions, such as electromagnetic noise, wide-ranging temperatures, high humidity, and vibration.

The IE340 Series provides network infrastructure for many vertical markets and related applications, such as:

- **Building automation**
Facility management including security and access control, fire protection, energy management, heating/ventilation/air-conditioning, and lighting control.
- **Cranes & Logistics**
Control of automated stacker cranes and other devices that boost the efficiency of dynamic warehouse environments.
- **Industrial automation and process control**
Interconnection of machines, IoT devices, sensors, and more. Instant communications between systems and people enables improved efficiency and resilience in manufacturing environments.
- **Marine control and monitoring**
Seamless communication for vessels, including high speed light water craft, and offshore units.
- **Railway transportation (signalling and telecommunications)**
Control signaling and telecommunication for improved safety, risk management, operating efficiency, and signage.
- **Railway transportation (traffic control)**
Adaptive traffic control, telematics, and preventive maintenance.
- **Smart cities**
Public space and urban infrastructure that provides safety and security, parking management, environmental metering, lighting, and information kiosks.
- **Wastewater treatment**
Industrial sewage treatment plants for efficient and reliable water purification. Control systems ensure process optimization by intelligent control, regulation, and monitoring.

IT/OT convergence

Improve productivity and decision-making by integrating your operational technology (OT) and information technology (IT). Use the intelligence of Industry 4.0 to collect, analysis, and manage all your data in real time.

Network automation and orchestration

Powerful automation options include Allied Telesis Autonomous Management Framework™ Plus (AMF Plus) and open standard-based northbound API.

For easy integration into complex networks consisting of physical, virtual, and multi-vendor devices, the IE340 Series features:

- NETCONF/RESTCONF + YANG data modelling for network automation.
- OpenFlow v1.3 for Software Defined Networking (SDN) orchestration.



AlliedWare Plus
OPERATING SYSTEM

Key Features

- 100 Mbps¹ / 1Gbps uplink ports
- EMC for industrial plants
- AlliedWare Plus™ operating system
- Allied Telesis Autonomous Management Framework™ Plus (AMF Plus)
- NETCONF/RESTCONF with YANG data modelling
- OpenFlow v1.3 for SDN
- QoS with traffic shaping
- Efficient forwarding of multicast streams
- Routing capabilities (BGP, ECMP, OSPF, RIP, and static)
- Extensive features for cybersecurity and denial of service prevention
- Active Fiber Monitoring™ (AFM)
- High Availability networking (EPSRing™, ITU-T G.8032, MRP)
- Automation and control protocols (Modbus/TCP, PROFINET IO)
- Upstream Forwarding Only (UFO)
- IEEE 802.3at PoE+ sourcing (up to 30W)
- 240W PoE power budget with dynamic allocation
- Continuous PoE
- Extended operating temp range: -40° C to 75° C (tested @85° C)¹
- Fanless design
- Graceful thermal shutdown
- Protection circuits
- Alarm monitoring with trigger facility
- Redundant power inputs
- Conformal coating

¹ Not supported on the IE340L model

KEY FEATURES

Network Automation

AMF Plus is a suite of tools that provide centralized control and network automation, as well as visual intent-based network management. It has the intelligence to set-up, optimize, and maintain the network according to predefined goals and policies.

Powerful features like centralized management, auto backup, auto upgrade, auto provisioning and auto recovery enable plug-and-play networking and zero touch management.

Integration with our Vista Manager visual monitoring and management platform means AMF Plus² also provides intent-based features like:

- Health monitoring to easily investigate, analyze and improve overall network health.
- Smart ACLs to control and secure the resources that clients use in the network.
- Intent-based QoS to deal with network bandwidth contention.

AMF Plus is scalable and can be either deployed integrated into Allied Telesis equipment, or on multi-tenant cloud architecture.

Northbound Interfaces

Open standard-based interfaces allow for easy integration with existing management systems.

NETCONF/RESTCONF with YANG data modeling provides a standardized way to represent data and securely configure devices.

OpenFlow is a key technology for SDN orchestration. SDN controllers and other tools support automated behavior in a network, and allow for the execution of customized applications and services.

Micro-segmentation for Network Security

Micro-segmentation enhances converged IT/OT network security by reducing the number of entry points for attackers or intruders. Isolating applications, data, and endpoints hampers the ability of intruders or malware to move within the network.

SDN network orchestration enables self-learning Artificial Intelligence to adapt and propagate security policies to mitigate evolving cyber threats.

High Availability

EPSRing™ and ITU-T G.8032 ERPS enable a protected ring capable of recovery within as little as 50ms. These features are perfect for high performance and high availability.

High-availability automation networks are supported with Media Redundancy Protocol (MRP) as defined by IEC62439-2.

MRP used in ring networks allows up to 50 devices to have guaranteed and deterministic switchover behavior.

Spanning Tree protocols RSTP and MSTP, along with static LAGs and the dynamic Link Aggregation Control Protocol (LACP), support high availability in star network topologies.

² From AW+ 5.5.2-2 onwards, an AMF Plus license operating in the network provides all standard AMF network management and automation features, and also enables the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10.1 onwards).

Automation and Control Protocols

Automation and control protocols enable integration with OT supervisory and control systems.

PROFINET IO is a communication protocol for data exchange between I/O controllers, like SCADA and PLC, with I/O devices over Ethernet networks.

Supporting PROFINET certification, the IE340 Series has I/O device properties that provide diagnostic data.

The switches execute these communication channels:

- Standard TCP/IP (PROFINET NRT): suitable for non-deterministic functions such as parametrization, video/audio transmissions and data transfer to higher level IT systems.
- Real Time (PROFINET RT): TCP/IP layers are bypassed in order to have deterministic performance for automation applications.

Modbus/TCP is intended for supervision and control of automation equipment. It is a variant of the MODBUS protocol using TCP/IP for communications on Ethernet networks.

The IE340 Series supports read/write register access and heartbeat functionality for efficient process control of both SCADA and slave devices.

Precise Time Synchronization (IEEE 1588)

Measurement and automation systems involving multiple devices often require accurate timing in order to facilitate event synchronization and data correlation. The IEEE 1588 Precise Time Protocol is a fault tolerant method enabling clock synchronization in a distributed system that communicates using an Ethernet network; this deterministic communication network is designed to provide precise timing for automation applications and measurement systems.

The IE340 Series supports IEEE 1588 (PTPv2) as a Transparent Clock, and performs an active role in Ethernet networks to reduce the effects of link delay and residence time..

Quality of Service (QoS)

Comprehensive low-latency wire-speed QoS provides flow-based traffic management with full classification, prioritization, traffic shaping and min/max bandwidth profiles. Enjoy boosted network performance and guaranteed delivery of business-critical services and applications.

sFlow

sFlow is an industry-standard technology for monitoring high-speed switched networks. It provides complete visibility into network use, enabling performance optimization, usage accounting/billing, and defense against security threats. Sampled packets sent to a collector ensure it always has a real-time view of network traffic.

Active Fiber Monitoring (AFM)

Active Fiber Monitoring prevents eavesdropping on fiber communications by monitoring received optical power. If the switch detects an intruder, it can automatically shut down the port or transmit an alert.

VLAN Mirroring (RSPAN)

VLAN mirroring allows traffic from a port on a remote switch to be analyzed locally. Traffic being transmitted or received on the port is duplicated and sent across the network on a special VLAN.

VLAN Translation

VLAN Translation allows traffic arriving on a VLAN to be mapped to a different VLAN on the outgoing paired interface.

VLAN Access Control List (ACLs)

ACLs simplify access and traffic control across entire segments of the network. They can be applied to a VLAN as well as a specific port.

Upstream Forwarding Only (UFO)

UFO lets you manage which ports in a VLAN can communicate with each other, and which only have upstream access to services, for secure multi-user deployment.

Dynamic Host Configuration Protocol (DHCP) Snooping

The switch keeps a record of the IP addresses of the devices on its ports, including those addresses allocated by DHCP servers. IP source guard checks against this DHCP snooping database to ensure only clients with specific IP and/or MAC addresses can access the network.

DHCP snooping works with other features, like dynamic ARP inspection, to increase security in Layer 2 switched environments, and also provides a traceable history which meets the growing legal requirements placed on service providers.

Link Layer Discovery Protocol–Media Endpoint Discovery (LLDP–MED)

LLDP-MED extends LLDP basic network endpoint discovery and management functions. LLDP-MED allows for media endpoint specific messages, providing detailed information on power equipment, network policy, location discovery (for Emergency Call Services) and inventory.

Power over Ethernet Plus (PoE+)

With PoE, a separate power connection to media endpoints is not necessary.

PoE+ reduces costs and gives even greater flexibility with up to 30W per port. This enables connecting high-power devices such as advanced security camera, kiosks, POS terminals, Wi-Fi access points, and LED light fixtures.

You may configure the overall PoE power budget to match the real capabilities of the external Power Supply Unit (PSU).

The PoE power budget may be allocated automatically and dynamically, based on the current usage of each powered device.

If the devices connected to a switch require more power than the switch can deliver, the switch will deny power to some ports, according to the assigned priority.

Continuous PoE

Continuous PoE allows the switch to be restarted without affecting the supply of power to connected devices. Smart lighting, security cameras, and other PoE devices will continue to operate during a software upgrade on the switch.

Port Based DHCP IP Address Assignment

DHCP server port-based address allocation ensures a replacement device receives the same IP address - even though the client-identifier or client hardware address has changed.

The supports Industrial Automation and Control Systems (IACS), which are very sensitive to operation outages. When devices such as sensors and actuators fail, they must be replaced immediately.

Assigning the same IP address to the replaced device allows the OT supervisory system to take control and resume operation as quickly as possible, minimizing downtime.

Alarm Monitoring and Trigger facility

The IE340 Series alarm facility monitors the switch and responds to any problems.

Example of alarm events include:

- Main power supply failure
- Over-temperature
- Port link down
- Alarm Input
- System power budget exceeded
- PoE device exceeds port power budget

Triggers based on alarm changes provide a smart/friendly mechanism for automatic and timed management of your device by activating the execution of commands in response to certain events.

Alarm Input/Output

Alarm Input and Output responds to an event instantly and automatically with predefined actions. The 2-pin terminal blocks may be connected to sensors and actuator relays.

Alarm Input receives signals from external devices like motion sensors and magnets that trigger specific actions when something changes.

Alarm Output controls external devices like strobes and sirens when an event occurs.

Protection Circuits

Optimized protection circuits guard against the following abnormal conditions:

- Reverse input voltage polarity
- Over- and under-voltage
- Over-current, peak-current and short-circuit
- Over-temperature

Enhanced Thermal Shutdown

Enhanced thermal shutdown acts to restrict PoE power and services when the switch exceeds a safe operating temperature.

The system restores operation when the temperature returns to acceptable levels.

Dual power inputs

The redundant power inputs are for higher system reliability and to allow UPS emergency power over an extended period of time.

Gigabit and Fast Ethernet SFP ports

The IE340 Series SFP ports support both Gigabit and Fast Ethernet Small Form-Factor Pluggables (SFPs).

This supports phasing in Gigabit fiber switches over time, allowing connectivity to legacy 100FX devices in the meantime. Supporting both speeds of SFP allows organizations to stay within budget as they migrate to faster technologies.

The IE340L supports Gigabit SFPs only.

Conformal coating

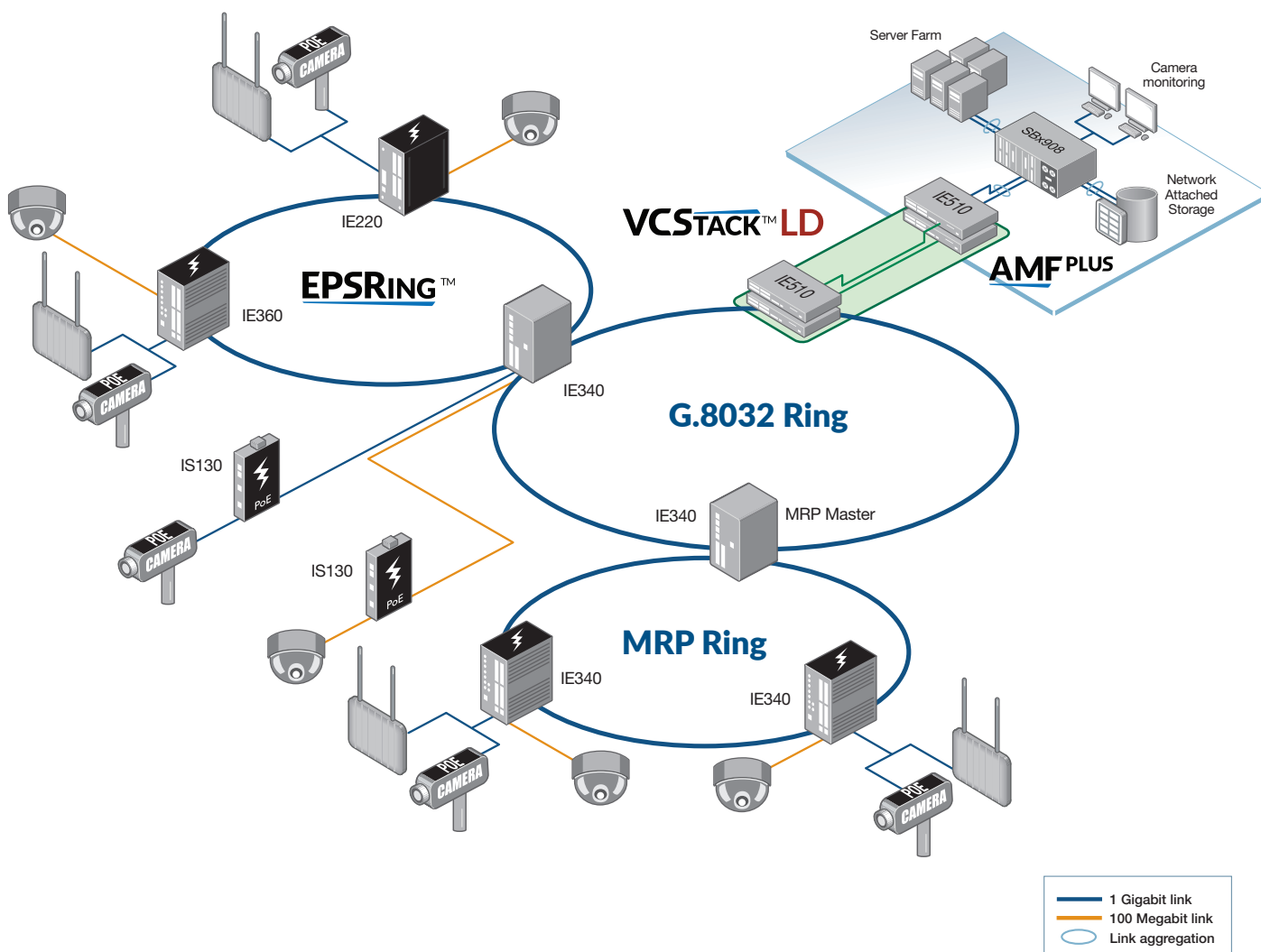
The IE340 Series may be protected by conformal coating, which is a protective chemical coating or polymer film that 'conforms' to the circuit board topology. Its purpose is to create a barrier to air-borne contaminants from the operating environment thus preventing corrosion from protecting electronic circuits.

Some typical application examples include wastewater treatment and marine, where coatings are used to protect against various combinations of moisture, aggressive chemicals and vapors, salt sprays, large temperature variations, mechanical vibration, and even organic attack (e.g., fungus).

Premium Software License

By default, the IE340 Series offers a comprehensive feature set that includes static routing. The feature set can easily be upgraded with premium software licenses.

Industrial Network



Media Redundancy Protocol (MRP), EPSRing and ERPS (ITU G.8032) provide high-speed resilient ring connectivity. This diagram shows how the IE Series can support a variety of ring network topologies.

The IE Series operates at a wide temperature range, and allows deployment in outdoor and harsh industrial environments.

PoE sourcing models support remotely controlled Pan, Tilt and Zoom (PTZ) video cameras, WiFi access points and more.

Management can be automated either with the Allied Telesis Autonomous Management Framework™ Plus (AMF Plus), or by 3rd party tools via the open standard northbound interface.

SPECIFICATIONS

Product Specifications

	10/100/1000T (RJ-45) Copper Ports	100M/1000X SFP Ports	1000X SFP Ports	Total Ports	PoE Enabled Ports	Switching Fabric	Forwarding Rate
IE340-12GP	8	4	-	12	8	24Gbps	17.8Mpps
IE340-12GT	8	4	-	12	-	24Gbps	17.8Mpps
IE340-20GP	16	4	-	20	16	40Gbps	29.7Mpps
IE340L-18GP	16	-	2	18	16	36Gbps	26.7Mpps

Physical Specifications

	Width	Depth	Height	Weight	Enclosure	Mounting	Protection Rate
IE340-12GP	91 mm (3.58 in)	139 mm (5.47 in)	153 mm (6.027 in)	DIN rail: 2.34 kg (5.16 lbs) Wall mount: 2.23 kg (4.91 lbs)	Aluminum/ Sheet Metal shell	DIN rail, wall mount	IP30
IE340-12GT							
IE340-20GP							
IE340L-18GP							

Power Characteristics

	Input Voltage ³	Cooling	No PoE Load			Full PoE+ Load ⁴		
			Max Power Consumption	Max Heat Dissipation	Noise	Max Power Consumption	Max Heat Dissipation	Noise
IE340-12GP	18~57V DC	Fanless	24W	81.9 BTU/h	--	271W	105.8 BTU/h	--
IE340-12GT						-	-	
IE340-20GP						271W	105.8 BTU/h	
IE340L-18GP	46~57V DC					271W	105.8 BTU/h	

³ PoE sourcing equipment require:
48Vdc to enable IEEE802.3at Type 1 (PoE)
54Vdc to enable IEEE802.3at Type 2 (PoE+)

⁴ The Max Power consumption at full PoE load includes the powered device's consumption and margin. The cooling requirements of the switch are smaller than the power draw, because most of the load is dissipated at the PoE powered device and along the cabling. Use these wattage and BTU ratings for facility capacity planning.

Power over Ethernet Sourcing Characteristics

	Enabled PoE Ports		Max PoE Power Budget ⁵	Max PoE Sourcing Ports	
	PoE	PoE+		PoE (15W)	PoE+ (30W)
IE340-12GP	8	8	240W	8	8
IE340-12GT	-	-	-	-	-
IE340-20GP	16	16	240W	16	8
IE340L-18GP	16	16	240W	16	8

⁵ The PoE power budget is shared among all ports; we recommend configuring dynamic PoE power allocation to optimize the power distribution

Latency (microseconds)

	Port Speed	
	100Mbps	1Gbps
IE340-12GP	10.98μs	2.44μs
IE340-12GT	11.02μs	2.22μs
IE340-20GP	10.98μs	2.44μs
IE340L-18GP	11.02μs	2.22μs

Performance

RAM memory	512MB DDR SDRAM
ROM memory	128MB flash
MAC address	16K entries
Packet Buffer	1.5 MBytes (16 Mbits)
Priority Queues	8
Simultaneous VLANs	4K
VLAN ID range	1–4094
Jumbo frames	9KB L2 jumbo frames
Multicast groups	511 (Layer 2), or 256 (Layer 2) and 256 (Layer 3) ⁶

⁶ When PIM is enabled.

Other Interfaces

Type	Serial console (UART)
Port no.	1
Connector	RJ-45 female
Type	USB2.0 (Host Controller Class)
Port no.	1 ⁶
Connector	Type A receptacle
Type	Alarm input (320μA @3.3Vdc)
Port no.	1
Connector	2-pin Terminal Block
Type	Alarm output (1A @30Vdc)
Port no.	1
Connector	2-pin Terminal Block

Flexibility and Compatibility

- SFP ports support any combination of Allied Telesis 100Mbps and 1Gbps SFP modules listed in this document under Ordering Information

Reliability

- Modular AlliedWare™ operating system
- Protection circuits against abnormal operations
- Redundant power input
- Full environmental monitoring of temperature and internal voltage levels
- Enhanced Thermal Shutdown
- MTBF (Mean Time Between Failures)
 - IE340-12GT: 570,000 hours
 - IE340-12GP: 500,000 hours
 - IE340-18GP: 430,000 hours
 - IE340-20GP: 390,000 hours

Industrial Automation

- IEEE 1588 PTP one-step variant
- IEEE 1588 PTP two-step variant
- IEEE 1588 PTP End-to-End Transparent Clock
- IEEE 1588 PTP Peer-to-Peer Transparent Clock
- IEEE 1588 PTP profile: Default
- Modbus/TCP with master/slave heartbeats facility
- PROFINET IO non-real-time and real-time (NRT/RT)

Management Features

- Allied Telesis Autonomous Management Framework™ Plus (AMF Plus) node
- NETCONF/RESTCONF northbound interface with YANG data modelling
- OpenFlow northbound interface
- Web-based Graphical User Interface (GUI)
- Industry-standard CLI with context-sensitive help
- Powerful CLI scripting engine
- Built-in text editor
- Event-based triggers allow user-defined scripts to be executed upon selected system events
- Link Layer Discovery Protocol (LLDP)
- Link Layer Discovery Protocol - Media Endpoint Discovery (LLDP-MED)
- SNMPv1/v2c/v3 support
- Comprehensive SNMP MIB support for standard based device management
- Console management port on the front panel for ease of access
- Front panel LEDs provide at-a-glance PSU status, PoE status, and fault information
- Eco-friendly mode allows ports and LEDs to be disabled to save power
- USB interface allows software release files, configurations, and other files to be stored for backup and distribution to other devices
- Recessed Reset button

IPv4 Features

- Black hole routing
- Directed broadcast forwarding
- Equal Cost Multi Path (ECMP) routing
- Dynamic routing (OSPF, RIP, and BGP)
- Static unicast and multicast routes for IPv4
- UDP broadcast helper (IP helper)

IPv6 Features

- Device management over IPv6 networks with SNMP, Telnet and SSH
- IPv4 and IPv6 dual stack
- IPv6 hardware ACLs
- Dynamic routing (OSPFv3, RIPng, and BGP+)
- Static unicast routing for IPv6
- IPv6 Ready certified

Multicasting Features

- Internet Group Management Protocol (IGMPv1/v2/v3)
- IGMP snooping with fast leave
- IGMP query solicitation
- Multicast Listener Discovery (MLDv1/v2)
- MLDv2 for IPv6
- MLD snooping
- IGMP/MLD proxy (multicast forwarding)
- Protocol Independent Multicast - Dense Mode (PIM-DM)
- Protocol Independent Multicast - Sparse Mode (PIM-SM)

Quality of Service

- 8 priority queues with a hierarchy of high priority queues for real-time traffic, and mixed scheduling, for each switch port
- Extensive remarking capabilities
- IP precedence and DiffServ marking based on Layer 2, 3 and 4 headers
- Limit bandwidth per port or per traffic class down to 64kbps
- Policy-based QoS and traffic shaping
- Policy-based QoS based on VLAN, port, MAC and general packet classifiers
- Policy-based storm protection
- Strict priority, weighted round robin or mixed scheduling
- Taildrop for queue congestion control
- Wirespeed traffic classification with low latency for real-time streaming media applications

Resiliency Features

- Control Plane Prioritization (CPP) ensures the CPU always has sufficient bandwidth to process network control traffic
- Dynamic link failover (host attach)
- Ethernet Protection Switching Ring (EPSR) with SuperLoop Prevention (EPSR-SLP)
- Ethernet Ring Protection Switching (ITU-T G.8032 ERPS)
- Link Aggregation Control Protocol (LACP)
- Loop detection and thrash limiting
- Media Redundancy Protocol (MRP)
- Multiple Spanning Tree Protocol (MSTP)
- PVST+ compatibility mode
- Rapid Spanning Tree Protocol (RSTP)
- Router Redundancy Protocol (RRP) snooping
- Spanning Tree Protocol (STP) root guard
- Virtual Router Redundancy Protocol (VRRPv3)

Security Features

- Access Control Lists (ACLs) based on layer 3 and 4 headers
- Authentication, Authorization and Accounting (AAA)
- Auth-fail and guest VLANs
- Configurable ACLs for management traffic
- BPDU protection

- DHCP snooping, IP source guard and Dynamic ARP Inspection (DAI)
- DoS attack blocking and virus throttling
- Dynamic VLAN assignment
- HTTP over TLS (HTTPS)
- MAC address filtering and MAC address lockdown
- Network Access and Control (NAC) features manage endpoint security
- Password protected bootloader
- Port-based learn limits (intrusion detection)
- Private VLANs and port isolation for multiple customers using the same VLAN
- RADIUS local server (100 users) and accounting
- Secure Copy (SCP)
- Strong password security and encryption
- Simple Certificate Enrollment Protocol (SCEP) supports secure management
- TACACS+ authentication and accounting
- Tri-authentication: MAC-based, web-based and IEEE 802.1X

Virtual LAN Features

- Generic VLAN Registration Protocol (GVRP)
- VLAN stacking, Q-in-Q
- VLAN translation
- Upstream Forwarding Only (UFO)

Services

- Domain Name System (DNS) client and relay
- Dynamic Host Configuration Protocol (DHCP) server and relay
- DHCPv6 server and relay
- HyperText Transfer Protocol (HTTP/1.1)
- Network Time Protocol (NTP) for IPv4 and IPv6
- Simple Mail Transfer Protocol (SMTP)
- Secure Shell (SSHv2/v3)
- TELNET
- Trivial File Transfer Protocol (TFTP)

Diagnostic Tools

- Active Fiber Monitoring (AFM) detects tampering on optical links
- Automatic link flap detection and port shutdown
- Built-In Self-Test (BIST)
- Cable fault locator (TDR)
- Connectivity Fault Management (CFM), Continuity Check Protocol (CCP) for use with ITU-T G.8032 ERPS
- Event logging via Syslog over IPv4
- Find-me device locator
- Optical Digital Diagnostic Monitoring (DDM)
- Ping polling for IPv4 and IPv6
- Port mirroring
 - No limit on mirrored ports
 - Up to 4 mirror (analyzer) ports for received traffic
 - 1 mirror (analyzer) port for transmitted traffic
- VLAN mirroring (RSPAN)
- sFlow
- TraceRoute for IPv4 and IPv6
- UniDirectional Link Detection (UDLD)

⁷ Coming in a later firmware release. Contact Sales representative for availability

Environmental Specifications

- Operating temperature range:⁸
IE340 model:
-40°C to 75°C (-40°F to 167°F)⁹
IE340L model:
-40°C to 65°C (-40°F to 149°F)
- Storage temperature range:
-40°C to 85°C (-40°F to 185°F)
- Operating humidity range:
5% to 95% non-condensing
- Storage humidity range:
5% to 95% non-condensing
- Operating altitude:
3,000 meters maximum (9,843 ft)

Mechanical

- EN 50022, EN 60715 standardized mounting on rails

Warranty

- Refer to the Term & Policies page on the Allied Telesis web site.

Compliance	IE340	IE340L
Compliance Mark	CE, FCC, ICES, IPv6 Ready, RCM, TEC, UKCA, UL, VCCI	
Hazardous Substances Compliance	RoHS, China-RoHS, JGSSI, REACH, SCIP, TSCA, WEEE	
Safety ¹⁰	AS/NZS 62368-1 CAN/CSA C22.2 No.60950-22 CAN/CSA C22.2 No.61010-1 CAN/CSA C22.2 No.62368-1 EN/IEC/UL 60950-22 EN/IEC/UL 61010-1 EN/IEC/UL 61010-2-201 EN/IEC/UL 62368-1	CAN/CSA C22.2 No.60950-22 CAN/CSA C22.2 No.62368-1 EN/IEC/UL 60950-22 EN/IEC/UL 62368-1
Electromagnetic Immunity	EN 55035 EN 61000-6-2	
Harmonic current emission	EN/IEC 61000-3-2 ¹⁰	
Voltage fluctuation and flicker	EN/IEC 61000-3-3 ¹⁰	
Electrostatic discharge (ESD)	EN/IEC 61000-4-2, level 3	
Radiated susceptibility (RS)	EN/IEC 61000-4-3, level 3, level x (20V/m)	
Electrical fast transient (EFT)	EN/IEC 61000-4-4,	Signal port: level 4 DC power port: level 3 Earth port: level 2
Lighting/surge immunity (Surge)	EN/IEC 61000-4-5, installation class 3 for outdoor Signal port: level 3 (L-E) DC power port: level 3 (L-E, R-E), level 2 (L-L)	
Conducted immunity (CS)	EN/IEC 61000-4-6, level 3	
Magnetic field	EN/IEC 61000-4-8, level 4	
AC voltage dips and interruption	EN/IEC 61000-4-11 ¹⁰	
DC voltage dips and Interruption	EN/IEC 61000-4-29 ¹¹	
Electromagnetic Emissions	AS/NZS CISPR 32, class A CISPR 11, group 1, class A CISPR 32, class A EN 55032, class A EN 61000-6-4, class A FCC 47 CFR Part 15, subpart B, class A ICES 003 class A VCCI class A	
Industry		
Marine	DNV ¹¹	-
Measurement, control and laboratory use	EN/IEC 61326-1	-
PROFINET IO	PROFIBUS certificate no. Z14174, Z14199, Z14200 PROFINET IO device, Conformance Class B add-on features: MRP Client, MRP Manager	
Programmable controller	EN/IEC 61131-2	-
Railway applications	EN 50121-4 (S/T apparatus)	
Traffic controller assemblies	NEMA TS 2	
Environmental		
Corrosive	ANSI/ISA S71.04, severity class G3 ⁸	
Freefall	IEC60068-2-31, class T2.3	
Shock	IEC60068-2-27 operational: 20g, 11ms, half-sine (DIN rail mount) 45g, 11ms, half-sine (wall mount) non-operational: 65g, 11ms, half-sine	
Vibration	IEC60068-2-6 operational: 2g @10~500Hz non-operational: 2g	

⁸ Refer to the Installation Guide for the full list of environmental tests.

⁹ Refer to the Installation Guide for more details on the safety approved power ratings and thermal conditions.

¹⁰ Test was applied using the power supply AT-IE048-480-20.

¹¹ Available on models with conformal coating.

STANDARDS & PROTOCOLS

AlliedWare Plus Operating System

Version 5.5.6-1

Authentication

RFC 1321	MD5 Message-Digest algorithm
RFC 1828	IP authentication using keyed MD5

Automation and Control

Modbus/TCP	
IEC 61158	Industrial communication networks - Fieldbus specifications - PROFINET
IEC 61784	Industrial communication networks - communication profile - PROFINET
IEEE 1588-2019	Precision Clock Synchronization Protocol

Border Gateway Protocol (BGP)

BGP dynamic capability	
BGP outbound route filtering	
RFC 1772	Application of the Border Gateway Protocol (BGP) in the Internet
RFC 1997	BGP communities attribute
RFC 2439	BGP route flap damping
RFC 2545	Use of BGP-4 multiprotocol extensions for IPv6 inter-domain routing
RFC 2918	Route refresh capability for BGP-4
RFC 3882	Configuring BGP to block Denial-of-Service (DoS) attacks
RFC 4271	Border Gateway Protocol 4 (BGP-4)
RFC 4360	BGP extended communities
RFC 4456	BGP route reflection - an alternative to full mesh iBGP
RFC 4724	BGP graceful restart
RFC 4760	Multiprotocol Extensions for BGP-4
RFC 5065	Autonomous system confederations for BGP
RFC 5492	Capabilities Advertisement with BGP-4
RFC 5925	The TCP Authentication Option
RFC 6793	BGP Support for Four-Octet Autonomous System (AS) Number Space
RFC 7606	Revised Error Handling for BGP UPDATE Messages

Encryption (Management Traffic Only)

FIPS 180-1	Secure Hash standard (SHA-1)
FIPS 186	Digital signature standard (RSA)
FIPS 46-3	Data Encryption Standard (DES and 3DES)

Ethernet

IEEE 802.2	Logical Link Control (LLC)
IEEE 802.3	Ethernet
IEEE 802.3ab	1000BASE-T
IEEE 802.3af	Power over Ethernet (PoE)
IEEE 802.3at	Power over Ethernet up to 30W (PoE+)
IEEE 802.3az	Energy Efficient Ethernet (EEE)
IEEE 802.3u	100BASE-X
IEEE 802.3x	Flow control - full-duplex operation
IEEE 802.3z	1000BASE-X

IPv4 Features

RFC 768	User Datagram Protocol (UDP)
RFC 791	Internet Protocol (IP)
RFC 792	Internet Control Message Protocol (ICMP)
RFC 793	Transmission Control Protocol (TCP)
RFC 826	Address Resolution Protocol (ARP)
RFC 894	Standard for the transmission of IP datagrams over Ethernet networks
RFC 919	Broadcasting Internet datagrams
RFC 922	Broadcasting Internet datagrams in the presence of subnets
RFC 932	Subnetwork addressing scheme
RFC 950	Internet standard subnetting procedure
RFC 951	Bootstrap Protocol (BootP)
RFC 1027	Proxy ARP
RFC 1035	DNS client
RFC 1042	Standard for the transmission of IP datagrams over IEEE 802 networks
RFC 1071	Computing the Internet checksum
RFC 1122	Internet host requirements
RFC 1191	Path MTU discovery
RFC 1256	ICMP router discovery messages

RFC 1518	An architecture for IP address allocation with CIDR
RFC 1519	Classless Inter-Domain Routing (CIDR)
RFC 1542	Clarifications and extensions for BootP
RFC 1591	Domain Name System (DNS)
RFC 1812	Requirements for IPv4 routers
RFC 1918	IP addressing
RFC 2581	TCP congestion control

IPv6 Features

RFC 1981	Path MTU discovery for IPv6
RFC 2460	IPv6 specification
RFC 2464	Transmission of IPv6 packets over Ethernet networks
RFC 3484	Default address selection for IPv6
RFC 3587	IPv6 global unicast address format
RFC 3596	DNS extensions to support IPv6
RFC 4007	IPv6 scoped address architecture
RFC 4193	Unique local IPv6 unicast addresses
RFC 4213	Transition mechanisms for IPv6 hosts and routers
RFC 4291	IPv6 addressing architecture
RFC 4443	Internet Control Message Protocol (ICMPv6)
RFC 4861	Neighbor discovery for IPv6
RFC 4862	IPv6 Stateless Address Auto-Configuration (SLAAC)
RFC 5014	IPv6 socket API for source address selection
RFC 5095	Deprecation of type 0 routing headers in IPv6
RFC 5175	IPv6 Router Advertisement (RA) flags option
RFC 6105	IPv6 Router Advertisement (RA) guard

Management

AT Enterprise MIB including AMF Plus MIB and traps	
Optical DDM MIB	
SNMPv1, v2c and v3	
ANSI/TIA-1057 Link Layer Discovery Protocol-Media Endpoint Discovery (LLDP-MED)	
IEEE 802.1AB Link Layer Discovery Protocol (LLDP)	
RFC 1155	Structure and identification of management information for TCP/IP-based Internets
RFC 1157	Simple Network Management Protocol (SNMP)
RFC 1212	Concise MIB definitions
RFC 1213	MIB for network management of TCP/IP-based Internets: MIB-II
RFC 1215	Convention for defining traps for use with the SNMP
RFC 1227	SNMP MUX protocol and MIB
RFC 1239	Standard MIB
RFC 1724	RIPv2 MIB extension
RFC 2011	SNMPv2 MIB for IP using SMIv2
RFC 2012	SNMPv2 MIB for TCP using SMIv2
RFC 2013	SNMPv2 MIB for UDP using SMIv2
RFC 2578	Structure of Management Information v2 (SMIv2)
RFC 2579	Textual conventions for SMIv2
RFC 2580	Conformance statements for SMIv2
RFC 2674	Definitions of managed objects for bridges with traffic classes, multicast filtering and VLAN extensions
RFC 2741	Agent extensibility (AgentX) protocol
RFC 2819	RMON MIB (groups 1,2,3 and 9)
RFC 2863	Interfaces group MIB
RFC 3176	sFlow: a method for monitoring traffic in switched and routed networks
RFC 3411	An architecture for describing SNMP management frameworks
RFC 3412	Message processing and dispatching for the SNMP
RFC 3413	SNMP applications
RFC 3414	User-based Security Model (USM) for SNMPv3
RFC 3415	View-based Access Control Model (VACM) for SNMP
RFC 3416	Version 2 of the protocol operations for the SNMP
RFC 3417	Transport mappings for the SNMP
RFC 3418	MIB for SNMP
RFC 3621	Power over Ethernet (PoE) MIB
RFC 3635	Definitions of managed objects for the Ethernet-like interface types
RFC 3636	IEEE 802.3 MAU MIB
RFC 4022	MIB for the Transmission Control Protocol (TCP)
RFC 4113	MIB for the User Datagram Protocol (UDP)
RFC 4188	Definitions of managed objects for bridges

RFC 4292	IP forwarding table MIB
RFC 4293	MIB for the Internet Protocol (IP)
RFC 4318	Definitions of managed objects for bridges with RSTP
RFC 4560	Definitions of managed objects for remote ping, traceroute and lookup operations
RFC 5424	The Syslog protocol
RFC 6527	Definitions of managed objects for VRRPv3

Multicast Support

Bootstrap Router (BSR) mechanism for PIM-SM	
IGMP query solicitation	
IGMP snooping (IGMPv1, v2 and v3)	
IGMP snooping fast-leave	
IGMP/MLD multicast forwarding (IGMP/MLD proxy)	
MLD snooping (MLDv1 and v2)	
PIM-SM and SSM for IPv6	
RFC 2236	Internet Group Management Protocol v2 (IGMPv2)
RFC 2710	Multicast Listener Discovery (MLD) for IPv6
RFC 2715	Interoperability rules for multicast routing protocols
RFC 3306	Unicast-prefix-based IPv6 multicast addresses
RFC 3376	IGMPv3
RFC 3590	Source Address Selection for the Multicast Listener Discovery (MLD) Protocol
RFC 3810	Multicast Listener Discovery v2 (MLDv2) for IPv6
RFC 3956	Embedding the Rendezvous Point (RP) address in an IPv6 multicast address
RFC 3973	PIM Dense Mode (DM)
RFC 4541	IGMP and MLD snooping switches
RFC 4604	Using IGMPv3 and MLDv2 for source-specific multicast
RFC 4607	Source-specific multicast for IP
RFC 7761	Protocol Independent Multicast - Sparse Mode (PIM-SM): Protocol specification

Open Shortest Path First (OSPF)

OSPF link-local signaling	
OSPF MD5 authentication	
OSPF restart signaling	
Out-of-band LSDB resync	
RFC 1245	OSPF protocol analysis
RFC 1246	Experience with the OSPF protocol
RFC 1370	Applicability statement for OSPF
RFC 1765	OSPF database overflow
RFC 2328	OSPFv2
RFC 2370	OSPF opaque LSA option
RFC 2740	OSPFv3 for IPv6
RFC 3101	OSPF Not-So-Stubby Area (NSSA) option
RFC 3509	Alternative implementations of OSPF area border routers
RFC 3623	Graceful OSPF restart
RFC 3630	Traffic engineering extensions to OSPF
RFC 4552	Authentication/confidentiality for OSPFv3
RFC 5329	Traffic engineering extensions to OSPFv3
RFC 5340	OSPFv3 for IPv6 (partial support)

Quality of Service (QoS)

IEEE 802.1p	Priority tagging
RFC 2211	Specification of the controlled-load network element service
RFC 2474	DiffServ precedence for eight queues/port
RFC 2475	DiffServ architecture
RFC 2597	DiffServ Assured Forwarding (AF)
RFC 2697	A single-rate three-color marker
RFC 2698	A two-rate three-color marker
RFC 3246	DiffServ Expedited Forwarding (EF)

Resiliency Features

IEC 62439-2	Media Redundancy Protocol (MRP)
IEEE 802.3ad	Static and dynamic link aggregation
EEE 802.1ag	CFM Continuity Check Protocol (CCP)
IEEE 802.1AX	Link aggregation (static and LACP)
IEEE 802.1D	MAC bridges
IEEE 802.1s	Multiple Spanning Tree Protocol (MSTP)
IEEE 802.1w	Rapid Spanning Tree Protocol (RSTP)
ITU-T G.8032 / Y.1344	Ethernet Ring Protection Switching (ERPS)
RFC 5798	Virtual Router Redundancy Protocol version 3 (VRRPv3) for IPv4 and IPv6

Routing Information Protocol (RIP)

RFC 1058	Routing Information Protocol (RIP)
RFC 2080	RIPng for IPv6
RFC 2081	RIPng protocol applicability statement
RFC 2082	RIP-2 MD5 authentication
RFC 2453	RIPv2

Security Features

SSH remote login	
SSLv2 and SSLv3	
TACACS+ Accounting, Authentication, Authorization (AAA)	
IEEE 802.1X	Authentication protocols (TLS, TTLS, PEAP and MD5)
IEEE 802.1X	Multi-supplicant authentication
IEEE 802.1X	Port-based network access control
RFC 2818	HTTP over TLS ("HTTPS")
RFC 2865	RADIUS authentication
RFC 2866	RADIUS accounting
RFC 2868	RADIUS attributes for tunnel protocol support
RFC 2986	PKCS #10: certification request syntax specification v1.7
RFC 3579	RADIUS support for Extensible Authentication Protocol (EAP)
RFC 3580	IEEE 802.1x RADIUS usage guidelines
RFC 3748	Extensible Authentication Protocol (EAP)
RFC 4251	Secure Shell (SSHv2) protocol architecture
RFC 4252	Secure Shell (SSHv2) authentication protocol
RFC 4253	Secure Shell (SSHv2) transport layer protocol
RFC 4254	Secure Shell (SSHv2) connection protocol
RFC 5176	RADIUS CoA (Change of Authorization)
RFC 5280	X.509 certificate and Certificate Revocation List (CRL) profile
RFC 5425	Transport Layer Security (TLS) transport mapping for Syslog
RFC 5656	Elliptic curve algorithm integration for SSH
RFC 6125	Domain-based application service identity within PKI using X.509 certificates with TLS
RFC 6614	Transport Layer Security (TLS) encryption for RADIUS
RFC 6668	SHA-2 data integrity verification for SSH
RFC 8446	Transport Layer Security (TLS) v1.3
RFC 8894	Simple Certificate Enrollment Protocol (SCEP)

Services

RFC 854	Telnet protocol specification
RFC 855	Telnet option specifications
RFC 857	Telnet echo option
RFC 858	Telnet suppress go ahead option
RFC 1091	Telnet terminal-type option
RFC 1350	The TFTP protocol (revision 2)
RFC 1985	SMTP service extension
RFC 2049	MIME
RFC 2131	DHCPv4 (server, relay and client)
RFC 2132	DHCP options and BootP vendor extensions
RFC 2616	Hypertext Transfer Protocol - HTTP/1.1
RFC 2821	Simple Mail Transfer Protocol (SMTP)
RFC 2822	Internet message format
RFC 3046	DHCP relay agent information option (DHCP option 82)
RFC 3315	Dynamic Host Configuration Protocol for IPv6 (DHCPv6)
RFC 3396	Encoding Long Options in the Dynamic Host Configuration Protocol (DHCPv4)
RFC 3633	IPv6 prefix options for DHCPv6
RFC 3646	DNS configuration options for DHCPv6
RFC 3993	Subscriber-ID suboption for DHCP relay agent option
RFC 4954	SMTP Service Extension for Authentication
RFC 5905	Network Time Protocol (NTP) version 4

VLAN LAN Features

Generic VLAN Registration Protocol (GVRP)	
Voice VLAN	
IEEE 802.1ad	Provider bridges (VLAN stacking, Q-in-Q)
IEEE 802.1Q	Virtual LAN (VLAN) bridges
IEEE 802.1v	VLAN classification by protocol and port
IEEE 802.3ac	VLAN tagging

FEATURE LICENSES

	Description	Includes
AT-IE340-FL01	IE340 Series Premium license	BGP (64 routes) BGP+ (64 routes) OSPF (64 routes) OSPFv3 (64 routes) PIM-SM, DM and SSM (256 routes) PIMv6-SM and SSM (256 routes) RIP (64 routes) RIPng (64 routes)

ORDERING INFORMATION

The DIN rail and wall mount kits are included. The management serial console cable is NOT included

AT-IE340-12GP-xx ¹²	8x 10/100/1000T, 4x 100/1000X SFP, Industrial Ethernet, Layer 3 Switch, PoE+ Support
AT-IE340-12GP/C-980	8x 10/100/1000T, 4x 100/1000X SFP, Industrial Ethernet, Layer 3 Switch, PoE+ Support, Conformal Coating
AT-IE340-12GT-xx ¹²	8x 10/100/1000T, 4x 100/1000X SFP, Industrial Ethernet, Layer 3 Switch
AT-IE340-20GP-xx ¹²	16x 10/100/1000T, 4x 100/1000X SFP, Industrial Ethernet, Layer 3 Switch, PoE+ Support
AT-IE340-20GP/C-980	16x 10/100/1000T, 4x 100/1000X SFP, Industrial Ethernet, Layer 3 Switch, PoE+ Support, Conformal Coating
AT-IE340-18GP-80	16x 10/100/1000T, 2x 1000X SFP, Industrial Ethernet, Layer 3 Switch, PoE+ Support

Where xx = 80 standard Country of Origin
980 TAA compliant Country of Origin

¹² Trade Act Agreement compliant (TAA)

Power Supplies	
AT-DRB50-48-1	50W @48Vdc, Industrial AC/DC power supply, DIN rail mount
AT-IE048-120-20	120W @48Vdc, Industrial AC/DC power supply, DIN rail mount (5 years warranty)
AT-IE048-240-20	240W @48Vdc, Industrial AC/DC power supply, DIN rail mount (5 years warranty)
AT-IE048-480-20	480W @48Vdc, Industrial AC/DC power supply, DIN rail mount (5 years warranty)
AT-IE048-480/C-20	480W @48Vdc, Industrial AC/DC power supply, DIN rail mount, Conformal Coating (5 years warranty)
AT-SDR120-48	120W @48Vdc, Industrial AC/DC power supply, DIN rail mount
AT-SDR240-48	240W @48Vdc, Industrial AC/DC power supply, DIN rail mount
AT-SDR480-48	480W @48Vdc, Industrial AC/DC power supply, DIN rail mount

Accessories

Refer to the installation guide for the recommended Maximum Operating Temperature according to the selected SFP module.

AT-VT-Kit3	Management cable (USB to serial console)
1000Mbps SFP Modules	
AT-SPBD10-13	10 km, 1G BiDi SFP, LC, SMF, I-Temp (1310 Tx/1490 Rx)
AT-SPBD10-14	10 km, 1G BiDi SFP, LC, SMF, I-Temp (1490 Tx/1310 Rx)
AT-SPBD20-13/I	20 km, 1G BiDi SFP, SC, SMF, I-Temp, (1310 Tx/1490 Rx)
AT-SPBD20-14/I	20 km, 1G BiDi SFP, SC, SMF, I-Temp, (1490 Tx/1310 Rx)
AT-SPBD20LC/I-13	20 km, 1G BiDi SFP, LC, SMF, I-Temp, TAA ¹³ (1310 Tx/1490 Rx)
AT-SPBD20LC/I-14	20 km, 1G BiDi SFP, LC, SMF, I-Temp, TAA ¹³ (1490 Tx/1310 Rx)
AT-SPBD40-13/I	40 km, 1G BiDi SFP, LC, SMF, I-Temp, (1310 Tx/1490 Rx)
AT-SPBD40-14/I	40 km, 1G BiDi SFP, LC, SMF, I-Temp, (1490 Tx/ 1310 Rx)
AT-SPEX/E-90	2 km, 1000EX SFP, LC, MMF, 1310 nm, Ext. Temp, TAA ¹²
AT-SPLX10a	10 km, 1000LX SFP, LC, SMF, 1310 nm, TAA ¹²
AT-SPLX10a/I	10 km, 1000LX SFP, LC, SMF, 1310 nm, I-Temp

AT-SPLX10/E-90	10 km, 1000LX SFP, LC, SMF, 1310 nm, Ext. Temp, TAA ¹²
AT-SPLX40	40 km, 1000LX SFP, LC, SMF, 1310 nm
AT-SPLX40/E-90	40 km, 1000LX SFP, LC, SMF, 1310 nm, Ext. Temp, TAA ¹²
AT-SPSX-90	550 m, 1000SX SFP, LC, MMF, 850 nm, TAA ¹²
AT-SPSX/I-90	550 m, 1000SX SFP, LC, MMF, 850 nm, I-Temp, TAA ¹²
AT-SPSX/E-90	550 m, 1000SX SFP, LC, MMF, 850 nm, Ext. Temp, TAA ¹²
AT-SPTX-90	100 m, 10/100/1000T SFP, RJ-45, TAA ¹²
AT-SPTX/I	100 m, 10/100/1000T SFP, RJ-45, I-Temp
AT-SPZX120/I	120 km, 1000LX SFP, LC, SMF, 1550 nm, I-Temp, TAA ¹²
100Mbps SFP Modules ¹³	
AT-SPFX/2-90	2 km, 100FX SFP, LC, MMF, 1310 nm, TAA ¹²
AT-SPFX30/I-90	30 km, 100FX SFP, LC, SMF, 1310 nm, I-Temp, TAA ¹²

¹² Trade Act Agreement compliant (TAA)

¹³ IE340L model does not support this feature)